UNIT

THREE

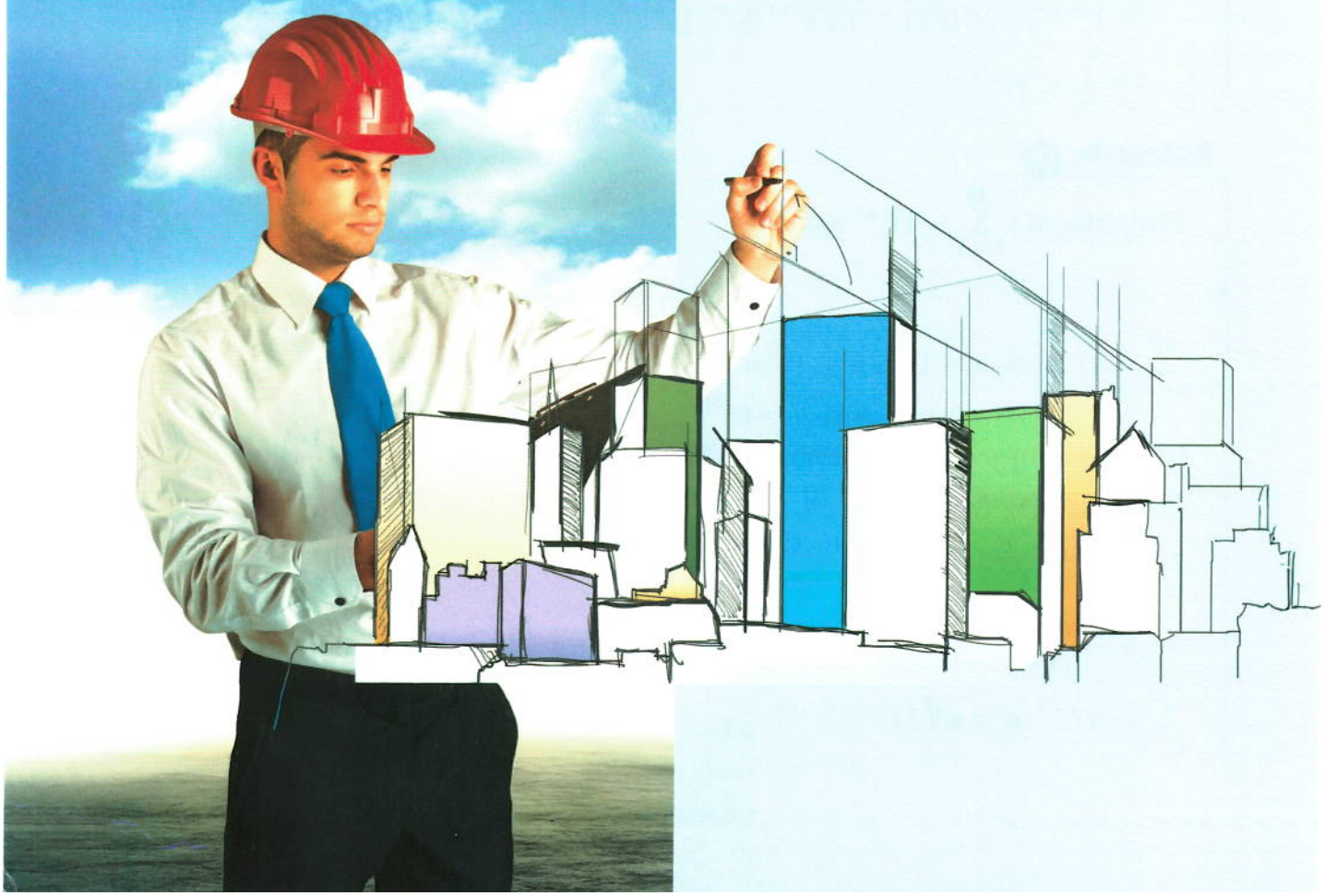
Geometry and measurement in three dimensions

Lesson 1 The 3-dimensional orthogonal coordinate system.

Lesson 2 Vectors in the space.

Lesson 3 Scalar product of two vectors.

Lesson 4 Vector product and scalar triple product.



Unit 3

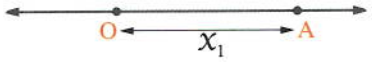
Lesson One

The 3-dimensional orthogonal coordinate system

• Determining the position of a body on a straight line (1-dimensional system)

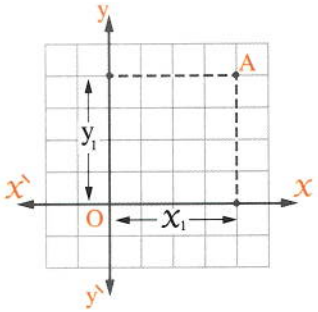
To determine the position of a body on a straight line , you should know the distance between the body and a fixed (arbitrary) point on it called the origin (O)

In the opposite figure : $OA = x_1 , x_1 \in \mathbb{R}$



• Determining the position of a body in a plane (2-dimensional system)

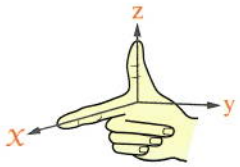
To determine the position of a body in a plane, you should know its projections on two dimensional orthogonal axis.



In the opposite figure : $A = (x_1 , y_1) \in \mathbb{R}^2$ where the straight line $\overleftrightarrow{xx}$ represents the x-axis, and the straight line $\overleftrightarrow{yy}$ represents the y-axis and the 2 axes are orthogonal and intersecting at the origin (O)

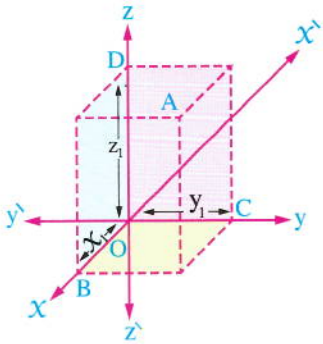
• Determining the position of a body in the space (3-dimensional system)

Assume 3 straight lines $\overleftrightarrow{xOx}$, $\overleftrightarrow{yOy}$, $\overleftrightarrow{zOz}$ in the space intersect at (O) and mutually orthogonal forming 3-dimensional orthogonal coordinate system according to the right hand rule as shown in the opposite figure.

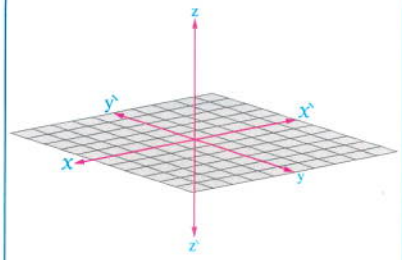
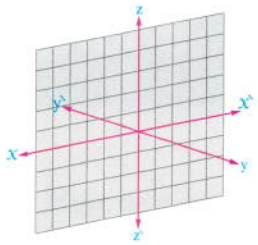
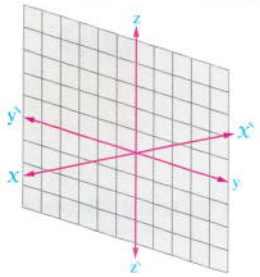


Where the index finger refers to the positive direction of x-axis , the middle finger refers to the positive direction of y-axis and the thumb refers to the positive direction of z-axis

The coordinates of the point A determined by the ordered triple $A (x_1 , y_1 , z_1) \in \mathbb{R}^3$ and the points $B (x_1 , 0 , 0)$, $C (0 , y_1 , 0)$ and $D (0 , 0 , z_1)$ are the projections of the point A on the 3 axes x , y and z respectively.



• **Cartesian planes :**

$\mathcal{X}$ y plane	$\mathcal{X}$ z plane	y z plane
		
It contains all points in the space whose coordinates are $(\mathcal{X}, y, 0)$ and its equation is $z = 0$	It contains all points in the space whose coordinates are $(\mathcal{X}, 0, z)$ and its equation is $y = 0$	It contains all points in the space whose coordinates are $(0, y, z)$ and its equation is $\mathcal{X} = 0$

Remark

Equation of the plane contains all the points in the form.

- $(\mathcal{X}, y, a)$ → $\mathcal{X}$ y – which is → $z = a$
- $(\mathcal{X}, a, z)$ → $\mathcal{X}$ z – which is → $y = a$
- (a, y, z) → y z – which is → $\mathcal{X} = a$

→ is the equation of the plane parallel to the plane.

→ Where a is constant $\in \mathbb{R}$

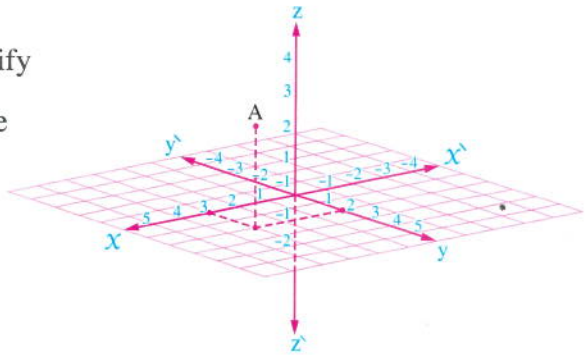
Example 1

Determine the position of each of the following points using 3D-orthogonal coordinate system :

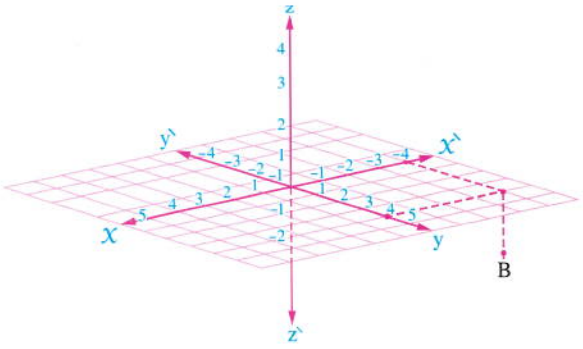
1. A $(3, 2, 3)$ **2.** B $(-4, 4, -2)$ **3.** C $(-2, 4, 0)$

Solution

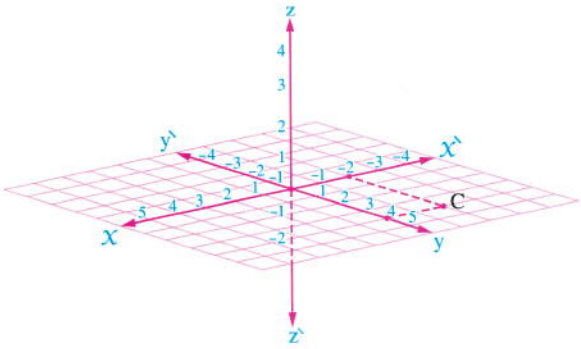
1. To determine the point A $(3, 2, 3)$, identify the point $(3, 2)$ in the $\mathcal{X}$ y plane then move 3 units in the positive direction of z-axis.



2.

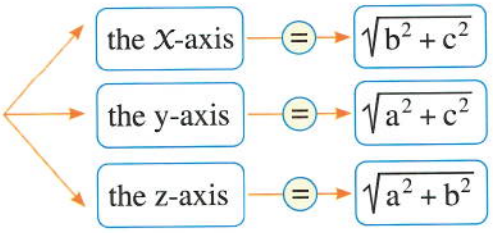


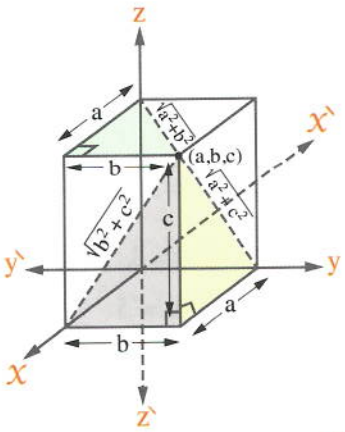
3.



The distance of a point from the coordinate axes

- The distance between the point (a , b , c) and





Remark

- The point (X , 0 , 0) lies on X-axis.
- The point (0 , y , 0) lies on y-axis.
- The point (0 , 0 , z) lies on z-axis.
- The point A (X_1 , y_1 , z_1)

→ Far from y z plane $|X_1|$ length unit.

→ Far from X z plane $|y_1|$ length unit.

→ Far from X y plane $|z_1|$ length unit.

For example : The point (5 , – 4 , 3) is far from y z plane 5 length unit , and far from X z plane 4 length units and far from X y plane 3 length unit.

- The two straight lines $\overleftrightarrow{XX}$, $\overleftrightarrow{yy}$ determine the X y plane , the two straight lines $\overleftrightarrow{XX}$, $\overleftrightarrow{zz}$ determine the X z plane , the two straight lines $\overleftrightarrow{yy}$, $\overleftrightarrow{zz}$ determine the y z plane.
- The equation of X-axis in the space is y = 0 , z = 0
- The equation of y-axis in the space is X = 0 , z = 0
- The equation of z-axis in the space is X = 0 , y = 0

Example 2

Find the coordinates of the point A in each of the following cases :

1. A $(2 + m , 3 - m , m) \in$ the $\mathcal{X}$ z plane.

2. A $(3\ell , \ell + 3 , 5\ell) \in$ the y-axis.

3. A $(2 + k , 4k , -k)$ at a distance 5 length units from y z plane.

4. A $(4m , 3m , m + 1)$ at a distance 5 units from z-axis.

Solution

1. $\because$ A $(2 + m , 3 - m , m) \in$ the $\mathcal{X}$ z plane

$\therefore 3 - m = 0$ $\therefore m = 3$ $\therefore$ A $(5 , 0 , 3)$

2. $\because$ A $(3\ell , \ell + 3 , 5\ell) \in$ the y-axis

$\therefore 3\ell = 5\ell = 0$ $\therefore \ell = 0$ $\therefore$ A $(0 , 3 , 0)$

3. $\because$ A $(2 + k , 4k , -k)$ at a distance 5 length units from y z plane

$\therefore 2 + k = 5$ $\therefore k = 3$ $\therefore$ A $(5 , 12 , -3)$

Or $2 + k = -5$ $\therefore k = -7$ $\therefore$ A $(-5 , -28 , 7)$

4. $\because \sqrt{(4m)^2 + (3m)^2} = 5$ $\therefore 16m^2 + 9m^2 = 25$

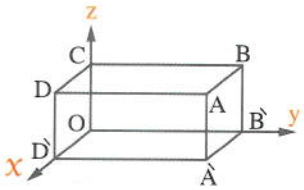
$\therefore 25m^2 = 25$ $\therefore m^2 = 1$ $\therefore m = \pm 1$

$\therefore$ A $(4 , 3 , 2)$ Or A $(-4 , -3 , 0)$

Example 3

In the opposite figure :

ABCD $\grave{A}$ $\grave{B}$ O$\grave{D}$ is a cuboid A $(3 , 7 , 5)$



Find :

1. The coordinates of the points B , $\grave{B}$, C , D , $\grave{D}$, $\grave{A}$

2. The volume of the cuboid.

Solution

1. B $(0 , 7 , 5)$ "Notice that B $\in$ the y z plane"

$\grave{B}$ $(0 , 7 , 0)$ "Notice that $\grave{B} \in$ the y-axis"

C $(0 , 0 , 5)$ "Notice that C $\in$ the z-axis"

D $(3 , 0 , 5)$ "Notice that D $\in$ the $\mathcal{X}$ z plane"

$\grave{D}$ $(3 , 0 , 0)$ "Notice that $\grave{D} \in$ the $\mathcal{X}$-axis"

$\grave{A}$ $(3 , 7 , 0)$ "Notice that $\grave{A} \in$ the $\mathcal{X}$ y plane"

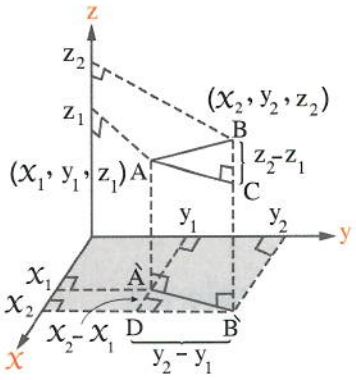
2. The volume of the cuboid $= 3 \times 7 \times 5 = 105$ cubic unit.

The distance between two points in the space

You have studied before the distance between the two points :

$A(x_1 , y_1)$, $B(x_2 , y_2)$ in the 2D-coordinate system where $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

But if the two points A , B are in the space where $A(x_1 , y_1 , z_1)$ and $B(x_2 , y_2 , z_2)$, then the distance between the two points A , B is given by the relation.



$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

For example : If A (2 , –3 , 5) , B (4 , 1 , –1) , then $AB = \sqrt{(2-4)^2 + (-3-1)^2 + (5+1)^2}$

$= 2\sqrt{14}$ length unit.

Example 4

If A (6 , 0 , 3) , B (7 , 1 , 7) , C (9 , 3 , 15) , prove that A , B , C are collinear.

Solution

$AB = \sqrt{(6-7)^2 + (0-1)^2 + (3-7)^2} = 3\sqrt{2}$ length unit.

, $BC = \sqrt{(7-9)^2 + (1-3)^2 + (7-15)^2} = 6\sqrt{2}$ length unit.

, $AC = \sqrt{(6-9)^2 + (0-3)^2 + (3-15)^2} = 9\sqrt{2}$ length unit.

$\therefore AC = AB + BC$ $\therefore$ A , B and C are collinear.

Example 5

Prove that Δ ABC where A (4 , 1 , 5) , B (– 4 , 3 , 1) , C (– 2 , 5 , 1) is obtuse-angled at C

Solution

$(AB)^2 = (4+4)^2 + (1-3)^2 + (5-1)^2 = 84$

, $(BC)^2 = (-4+2)^2 + (3-5)^2 + (1-1)^2 = 8$

, $(AC)^2 = (4+2)^2 + (1-5)^2 + (5-1)^2 = 68$

, $\because (AB)^2 > (AC)^2 + (BC)^2$

$\therefore$ Δ ABC is obtuse-angled at C

Remember

In Δ ABC , if :

- $(AB)^2 = (BC)^2 + (AC)^2$, then ∠ C is right.
- $(AB)^2 > (BC)^2 + (AC)^2$, then ∠ C is obtuse.
- $(AB)^2 < (BC)^2 + (AC)^2$, then ∠ C is acute.

The coordinates of the midpoint of a line segment

If A (x_1 , y_1 , z_1) , B (x_2 , y_2 , z_2) are two points in the space , then the point C which is the midpoint of $\overline{AB}$ is : $C = \left(\frac{x_1 + x_2}{2} , \frac{y_1 + y_2}{2} , \frac{z_1 + z_2}{2}\right)$

- **For example :** If A $(5 , -4 , 0)$, B $(1 , 8 , 4)$, then the midpoint of $\overline{AB} = (3 , 2 , 2)$

Example 6

If C $(3 , 2 , -1)$ is the midpoint of $\overline{AB}$ where A $(4 , 3 , -1)$, find the coordinates of B

Solution

Let the point B be (x , y , z)

, $\because$ C is the midpoint of $\overline{AB}$ $\quad\quad$ $\therefore (3 , 2 , -1) = \left(\frac{4 + x}{2} , \frac{3 + y}{2} , \frac{-1 + z}{2}\right)$

$\therefore 4 + x = 6$ $\quad\quad$ $\therefore x = 2$

$, 3 + y = 4$ $\quad\quad$ $\therefore y = 1$

$\therefore -1 + z = -2$ $\quad\quad$ $\therefore z = -1$

$\therefore$ B $(2 , 1 , -1)$